

Image Rotation (Clockwise & Counter-Clockwise)

Aim: To rotate an image in both clockwise and counter-clockwise directions using OpenCV in Python.

Procedure:

1. Import cv2.
2. Load the image using cv2.imread("path_to_image").
3. Rotate 90° clockwise using cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE).
4. Rotate 90° counter-clockwise using cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE).
5. Display all outputs using cv2.imshow() and close windows with cv2.destroyAllWindows().

```
import cv2
```

```
# Read the image
```

```
img = cv2.imread(r"C:/Users/MOSHIYA/OneDrive/Documents/Toy.png")
```

```
if img is None:
```

```
    print("Error: Could not load image. Check the path and file name.")
```

```
else:
```

```
    # 1. Rotate 90 degrees Clockwise
```

```
    img_clockwise = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)
```

```
    # 2. Rotate 90 degrees Counter-Clockwise
```

```
    img_counter_clockwise = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)
```

```
# Display images
```

```
cv2.imshow("Original", img)
```

```
cv2.imshow("Clockwise 90°", img_clockwise)
```

```
cv2.imshow("Counter-Clockwise 90°", img_counter_clockwise)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

